Fiscal Year:	FY 2010	**Task Last Updated:**	FY 10/30/2009
PI Name:	Redpath, John Leslie Ph.D.		
Project Title:	High energy proton dose-rate and mixed field effects on neoplastic transformation in vitro		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jlredpat@uci.edu	**Fax:**	FY (949) 824-3566
PI Organization Type:	UNIVERSITY	**Phone:**	(949) 824-7395
Organization Name:	University of California Irvine		
PI Address 1:	Radiation Oncology		
PI Address 2:	School of Medicine, Med Sci I, B148		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697	**Congressional District:**	48
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	01/01/2007	**End Date:**	12/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received extension to 12/31/2011 (from 12/31/2010) per C. Guidry/JSC (Ed., 2/3/11)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Elmore, Eugene (University of California Irvine)		
Grant/Contract No.:	NNJ06HD90G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The central objectives are 1) to determine the the factor by which the effectiveness of high energy protons is reduced when they are delivered at dose-rates that correspond to solar particle events, and 2) to determine the effectiveness of mixed fields of protons and heavy ions compared to either alone. The cancer-relevant endpoint of neoplastic transformation in vitro will be used. The work is of relevance to the improvement of estimation of cancer risks to astronauts involved in space exploration and address certain emphases in the Bioastronautics Roadmap.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Provides greater insight into cancer risk from exposure to low doses of high energy proton and iron ion irradiation.
Task Progress:	We have now completed the dose-response curve for radiation-induced neoplastic transformation for 1 GeV/n iron ions in the dose range 10 to 1000 mGy. The results indicate that the curve shows a threshold response with no induction of transformation at doses <300 mGy. This finding has held up over repeated assays. We hypothesize that this finding may be due to an adaptive response from the associated delta ray penumbra. This hypothesis could be tested by using particle radiation of similar LET but with a penumbra restricted to sub-cellular dimensions, e.g. alpha particles. A comparison of the 1 GeV/n iron dose-response curve to that for Cs-137 gamma rays reveals the iron to be more effective at inducing transformation at all doses tested. As we have seen previously, at doses <300 mGy Cs-137 gamma radiation suppresses transformation frequency below that seen in unirradiated controls. The RBE for iron compare to Cs-137 at doses >300 mGy is around 1.6. A paper on the above work was recently published (Radiation Research 17: 764-770, 2009). We have now performed three sets of mixed beam experiments at NSRL (Fall 2007, Summer 2008, Spring 2009). These experiments were to test the effect of 10 cGy of 1 Gev/n protons combined with 100 cGy of 1 GeV/n iron and 10cGy of 1 GeV/n iron followed by 100 cGy 1 GeV/n protons. In each case the second dose was delivered either immediately (<15 min), or after a 24 h delay, following the first dose. The hypothesis was that the proton irradiation could induce an adaptive response that would ameliorate the effectiveness of the high dose of iron, especially when the iron dose was delayed with respect to the proton dose. We have now run 4 repeat assays from these experiments and have found that the data do not agree with our original hypothesis. For low dose protons followed by high dose iron there was no evidence of an adaptive response with the results being statistically consistent with additivity. Surprisingly, for low dose iron followed bu high dose protons there was an apparent adaptive response. These assays were only completed in late September 2009 and we are in the process of preparing a manuscript as well as pondering additional studies to explain this unexpected effect. In addition to the mixed beam studies reported above we also performed dose-response curve experiments for 1 GeV/n protons and 1 GeV/n iron. These assays have also been run over the past year. The data for iron supported the previous work that we have recently published and extended the doses down to 0.1 cGy. The data for protons supported our previous finding with 250 MeV/n protons form the Loma Linda facility that we published several years ago in that a suppression of transformation frequency was evident at doses <10 cGy. The experiments to determine the dose and dose-rate effectiveness factor (DDREF) for high energy (250 MeV) protons continue at Loma Linda. In 2007 year we ran 4 experiments that gave equivocal results. In late 2008 and early 2009 we have repeated these experiments using the technique of post-irradiation freezing of the irradiated samples for assay at a later time, just as we do for the NSRL experiments. Now that we have completed the assays on the mixed beam experiments we are just starting to run these assays. It is anticipated that that task will take six more months to complete. Summary Our findings to date indicate that low doses (=10 cGy) of 1GeV iron ions may be less hazardous than previously thought with respect to cancer induction. This was a somewhat unexpected finding but does have a plausible explanation in terms of an adaptive response induced by the low-LET delta ray penumbra. The results with 1 GeV/n protons are consistent with previous findings showing a suppression of transformation at doses =10 cGy. The results with mixed beam exposures are also unexpected with a prior dose of 10 cGy protons having little or no impact on the effect of 100 cGy iron ions. On the other hand prior low doses of iron had an apparent protective effect on a subsequent 100 cGy dose of protons. Taken at face value these data would indicate that mixed beam exposures do not pose any additional hazard over each particle separately. Of course the critical experiments that more closely simulate space radiation will be the low dose-rate proton studies that have yet to be completed.
Bibliography Type:	Description: (Last Updated: 01/13/2012)
Articles in Peer-reviewed Journals	Elmore E, Lao X-Y, Kapadia R, Redpath JL. "Threshold-type dose response for induction of neoplastic transformation by 1 GeV/nucleon iron ions." Radiation Research. 2009 Jun:171(6):764-70. PMID: 19580483 , Jun-2009